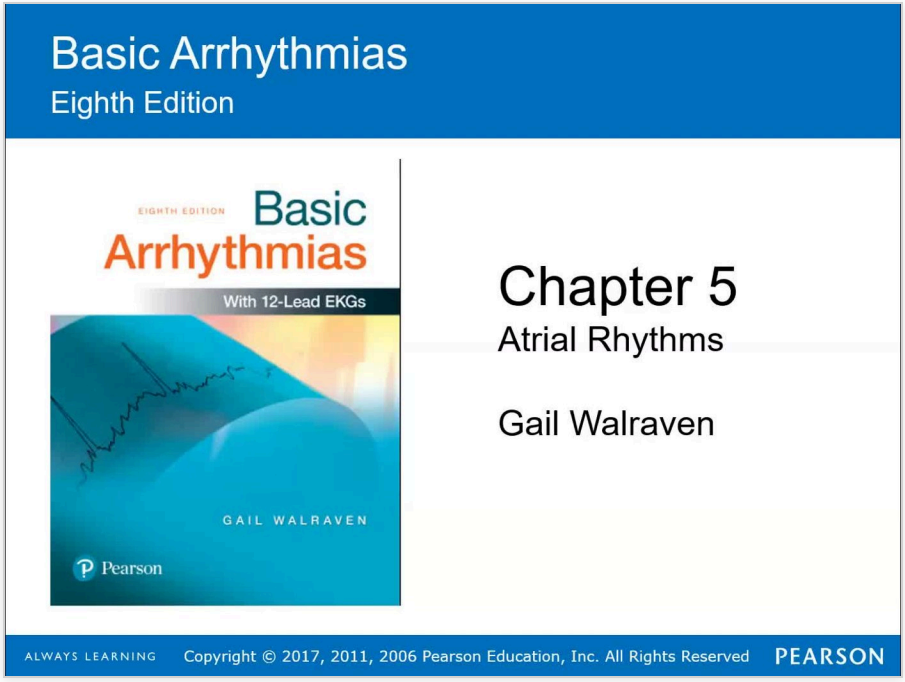
Lecture Transcript by Slide

Week 7 · Chapter 5: Atrial Rhythms · first 1 h 8 min of the Atrial and Junctional Rhythms recording
Auto-transcribed, so expect occasional errors, especially in technical terms. Timestamps are positions in the recording.

0:00:00	Chapter 5: Atrial Rhythms (title slide)
0:01:00	Wandering Pacemaker
0:04:50	Rules for Wandering Pacemaker
0:06:08	Wandering Pacemaker
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1:08:01	Chapter 5 Practice Strips

1 Chapter 5: Atrial Rhythms (title slide)

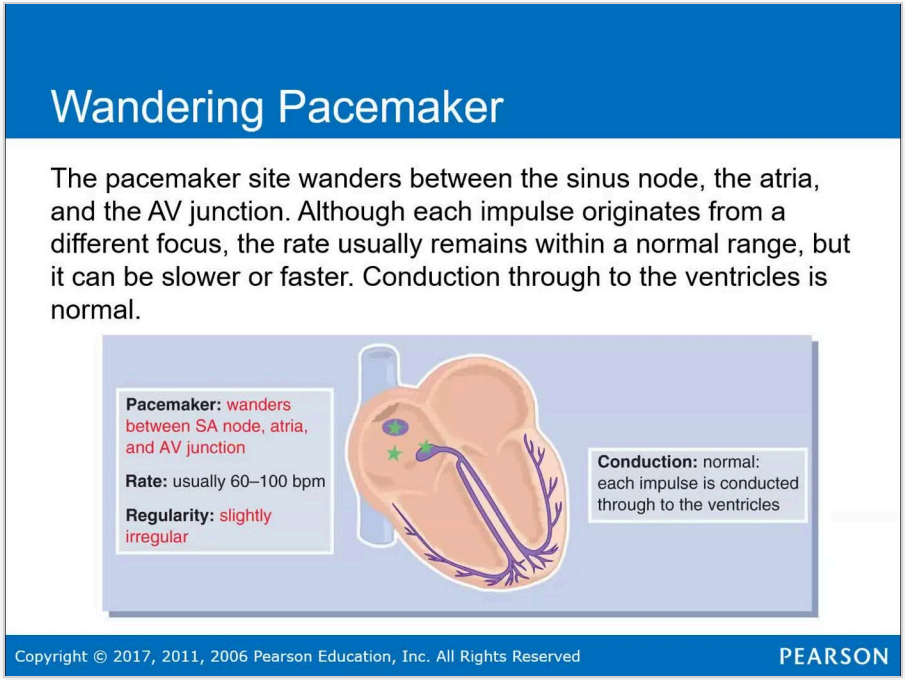
0:00:00 – 0:01:00



0:00:00 Okay. So these are going to be rhythms generated from any of the atria muscle cells. So the right or left atria, one of the cardiac muscle cells in the atria can generate single beats or can generate an entire rhythm.

2 Wandering Pacemaker

0:01:00 – 0:04:50



0:01:00 So the first rhythm that we see in the PowerPoint is really an odd rhythm called wandering pacemaker. This particular rhythm is extremely rare. I've been teaching this course for 25 years and I teach in the ICU and I don't think I've ever seen this rhythm clinically. So I'm going to teach it to you. There will be one strip on the homework assignment that will be interpreted as a wandering pacemaker, but this will not be on the test or the final exam. I'm not going to test you on something that is really rare and not common and generally not life threatening. If it was life threatening, I would probably include it, but it's not in any way, shape or form.

0:01:54 So this rhythm, this rhythm, its name really says what it is. So the pacemaker site, which is normally the SA node, wanders. So one beat will originate from one of the atrial muscle cells, like where the green star is. And then the next beat might come from the AV junction itself.

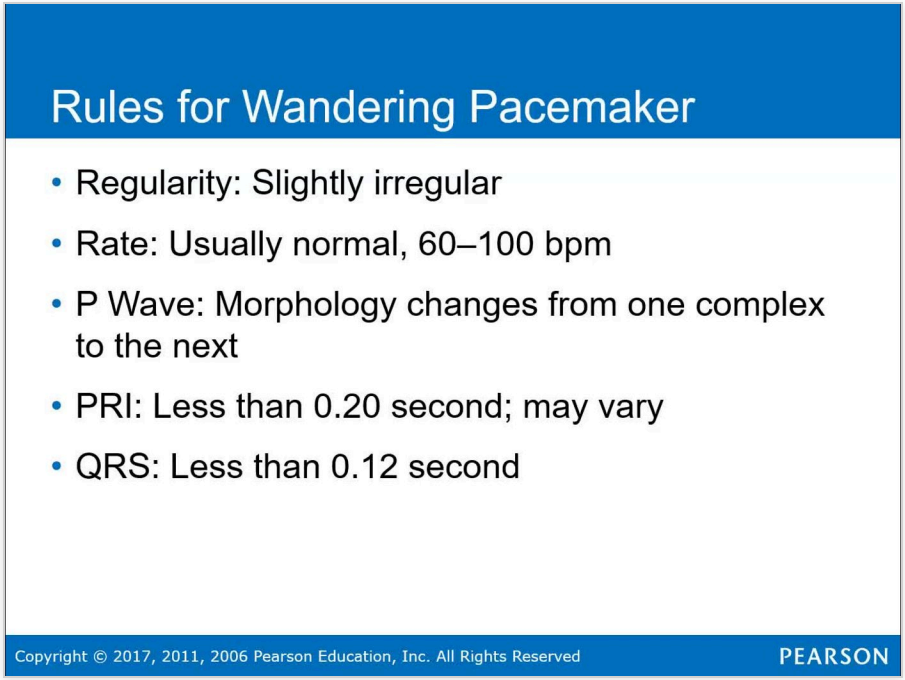
0:02:29 So it wanders from one pacemaker site to the other, to the other, to the other, to the other, to the other. Even though we have a wandering pacemaker coming from different focal points, like I said, the SA node, an atrial muscle cell or the AV node itself. So those are different focal points. So even though we have beats coming from a different pacemaker or focal point, the rate for this rhythm usually stays within the 60 to 100 beats per minute. Can be a little slower, a little faster. And then conduction through the ventricles is normal. So that means it's going to be narrow, 0.12 seconds or less.

0:03:23 All of these pacemaker sites, the SA node and atrial muscle cell and the AV junction are all supra-ventricular pacemaker cells. So in order to depolarize the ventricles, it has to go through the bundle of his, the bundle branches to the Purkinje fibers. Remember, when the impulse travels over the nerve fibers, it travels fast. It's delivered quickly and depolarization of the ventricles happens quickly, which is normal. Remember that EKGs are electrical activity recorded over time. So if conduction and depoloration is quick, the waveform will be narrow, not wide. We've talked about this over the last few classes.

0:04:12 So you can see where the green stars are, the SA node to the atrial muscle cell to the AV node itself, okay? Even though the rate stays the same, our R-to-R spacing might be slightly irregular because they don't necessarily fire at regular intervals, okay? And as I go through these rhythms, I will certainly try to identify what are the main features or the key features that you need to understand that will help you identify this rhythm as wandering pacemaker. And I'll do that when we get to the interpretation piece.

3 Rules for Wandering Pacemaker

0:04:50 – 0:06:08

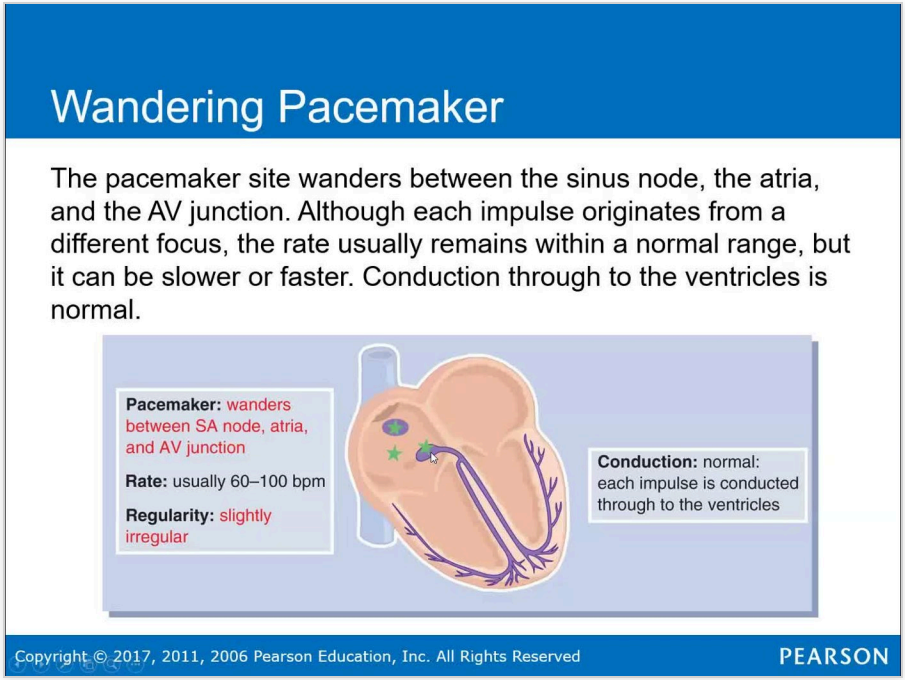


0:04:54 So here are basically the guidelines or the rules for wandering pacemaker. So R-to-R is slightly irregular. So remember, up to a box and a half, we still consider it regular. So we might get R-to-R that's one and a half, maybe one is two, maybe one is two and a half difference when you measure out the small boxes between R waves. So the R-to-R is going to be slightly irregular. The rate is usually normal. Now here's one of the keys to identifying wandering pacemaker. P-wave, morphology means shape.

0:05:34 So we know that sinus P-waves are round and dome-shaped. You're going to learn today that atrial P-waves can be notched, peaked, or flattened. They're not dome-shaped. And then P-waves coming from the junction are either A, not visible, or B, they're inverted or upside down. They're negatively deflected. Remember when we talked a decent amount about electrical flow going towards a negative electrode produces a negative waveform.

4 Wandering Pacemaker

0:06:08 – 0:07:28

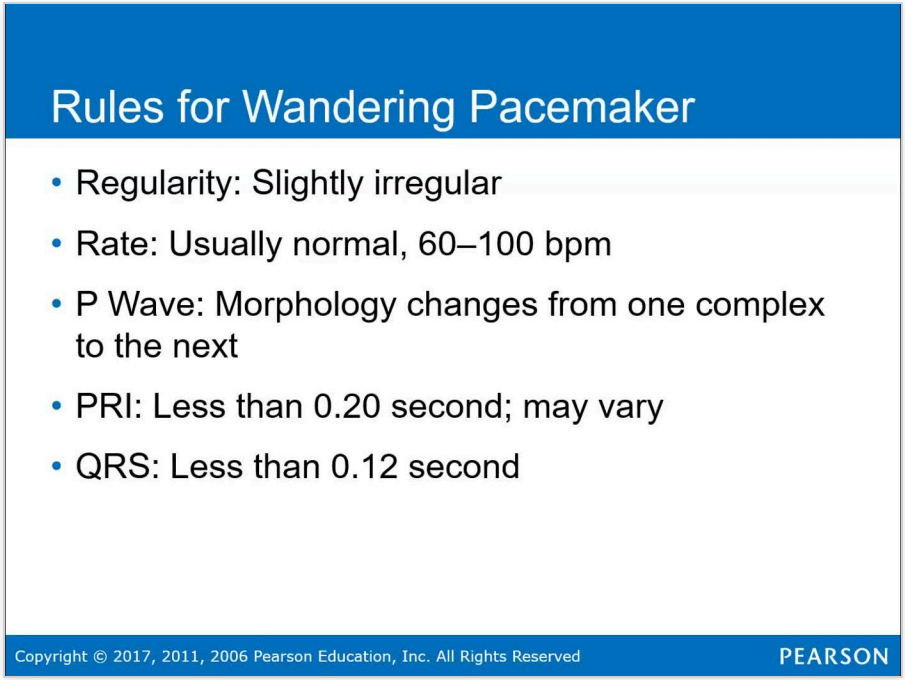


0:06:18 So if the AV node is the pacemaker, in order to depolarize the atrium muscle cells, electrical flow has to go back towards the right shoulder. The right shoulder is a negative electrode. So when we depolarize the atria from a AV node B, the P-waves will be inverted or upside down. So if I draw an isoelectric line, instead of the P-waves being upright, positively deflected, it's going to be the opposite. It's going to be upside down or negatively deflected. We call this an inverted P-wave. And you're going to learn about junctional P-waves today. That's going to be the second half of our lecture. Okay?

0:07:16 So the P-waves for the junctional beats are either not visible, and I'll explain why when we get to that point, or B, they will be inverted or upside down.

5 Rules for Wandering Pacemaker

0:07:28 – 0:08:23

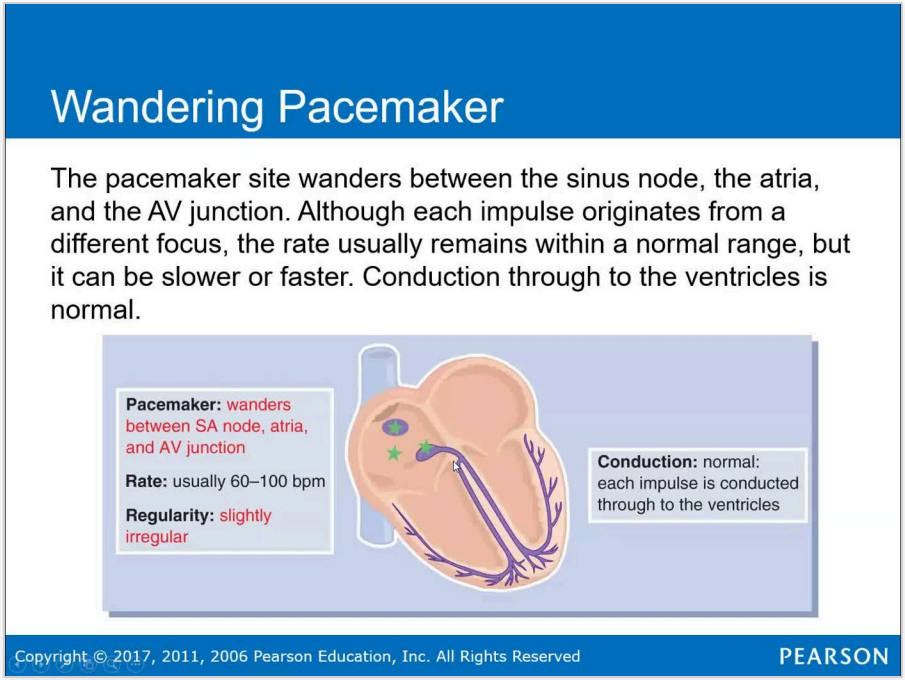


0:07:29 And remember, we're going from the SA node to an atrial muscle cell to an AV node or a junction B. So each one of these successive P-waves is going to look slightly different. They won't be upright and uniform like they are from the SA node. The SA nodes will be upright and rounded. The atrium muscle cells might be peaked, notched, or flattened. The junctional P-wave is either going to be not visible or it's going to be inverted. So what you'll see is a change in the shape of the P-waves. What I've taught you so far also is that the PRI measurement in sinus rhythms stays constant across the strip.

0:08:17 Well, if we have beats coming from different pacemakers, and if you look at the picture again, these pacemakers are different distances to the AV node.

6 Wandering Pacemaker

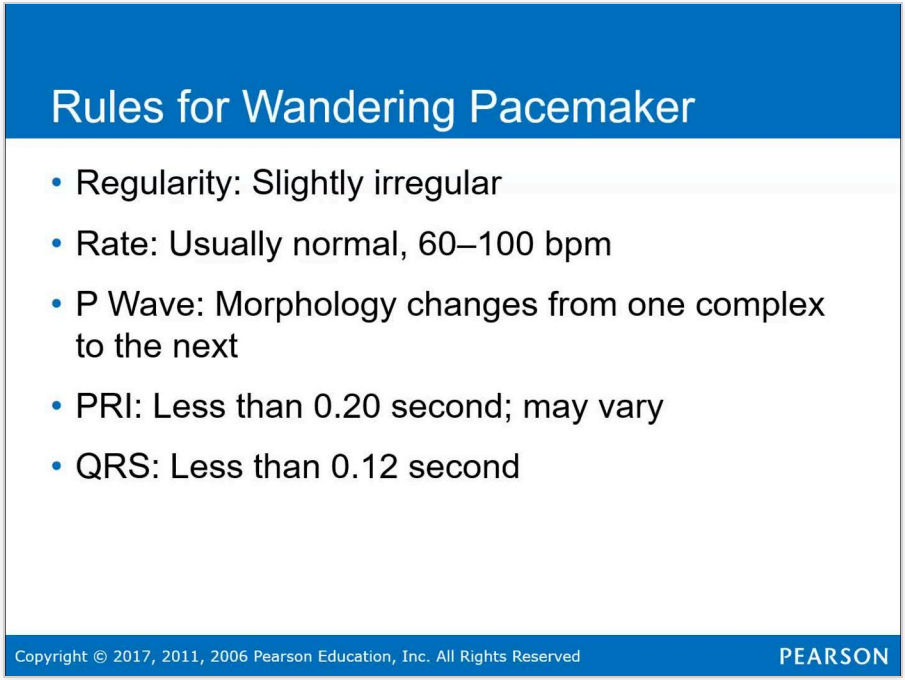
0:08:23 – 0:09:13



0:08:30 So the farther the pacemaker is away from the AV node, the longer the PRI will be. The closer it is to the AV node, the shorter the PRI. So not only will the P-waves change in shape, the PRI will change in distance or time. So like the SA node, PRI might be four boxes at 0.16, but the atrial cell is 0.12. So that one is one small box shorter. Or the AV node one has a PRI of 0.08. So, you know, when you're looking at these guidelines or rules, you need to put an asterisk next to the P-wave shape and the PRI.

7 Rules for Wandering Pacemaker

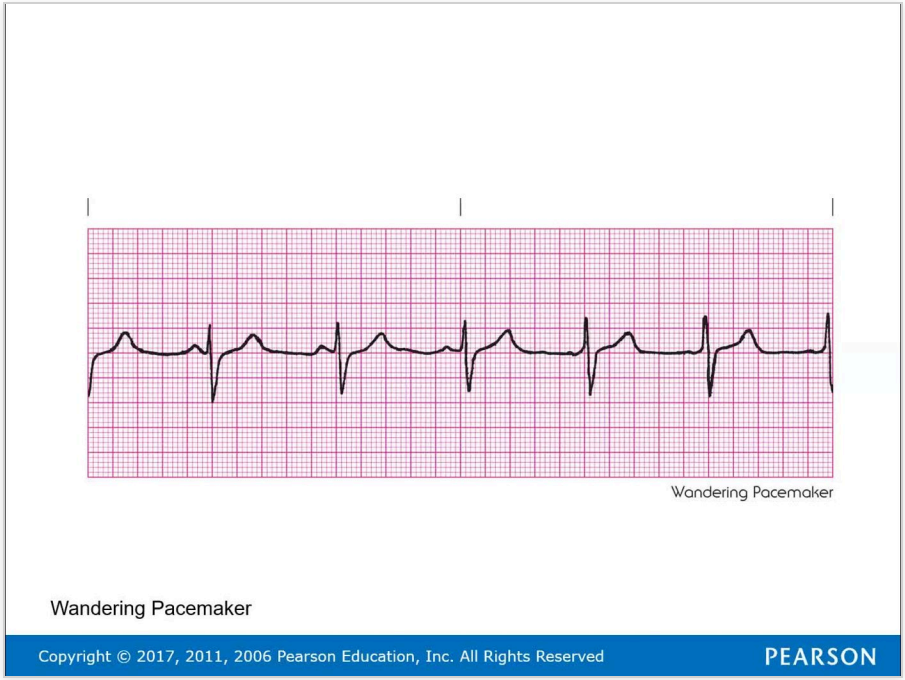
0:09:13 – 0:09:40



0:09:18 Those are the two key identifying features of this wandering pacemaker rhythm. And then we already talked about how the QRS will be normal or narrow at 0.12 seconds or less.

8 Wandering Pacemaker (ECG strip)

0:09:40 – 0:20:01



0:09:41 And this is what it looks like. So let's go through our information gathering and interpret this rhythm. So we can't really measure R to R at the beginning of this rhythm because we can't see the R-wave from this particular beat right here.

0:10:02 Well, we can measure from there to there. So 5, 10, 15, 20, 25. Yeah, we call that two halves. So 26 small boxes between the first and the second R-wave. 5, 10, 15, 20, 25. Maybe 25 and a half. Okay.

0:10:32 5, 10, 15, 20, 24. So we got 26, 25 and a half, 24. So already we have a rhythm that is outside the one and a half small box regular classification. Let's do the next one. 5, 10, 15, 20, 24. 5, 10, 15, 20, 24 and a half. So we go from 24 up to 26. Okay.

0:11:12 So what's fair for me to expect from you with this rhythm because we have a maximum of two box difference. And I told you greater than one and a half is basically irregular. So if you tell me under the regularity section for this, it's slightly irregular or you put irregular, I would take either one as the answer based on what I've told you. Or what I've taught you. So slightly irregular or irregular. How do we count the R's and multiply by 10? How many R's are here? Right. Because I would actually count the last one and the first one as one whole B. Because it's a half on both ends.

0:12:05 Right. So I agree. That's six. So the heart rate is 60. Now let's look at the P waves. So the first one. Kind of upright and dome shaped. The second one. Kind of looks similar to the first one. It's upright and dome shaped. The third one. Smaller. Less amplitude. The fourth one. Even smaller. The fifth one. Looks similar to the fourth one. And then the last one. Is probably somewhere in between.

0:12:50 The fourth and fifth. And the third. So you can clearly see that the P wave shape is different here. So what should you write under the P wave? I would just simply write shape changes. Now we've got to measure the PRI.

0:13:20 The first PRI looks like it's four small boxes. It starts right at the large box. And that R wave goes up right there. Right at that line. So I would call it four small boxes.

0:13:41 The next one I still think is four small boxes. It starts halfway in between right here where I have the cursor. And the R wave goes up halfway in between. So that's four small boxes apart. As I go through this, if you don't know what I'm talking about, you got to ask. You got to stop me and ask. The next one, that P wave starts just after that line. It's not on that line. It's after that line. And the R wave is right there. So I actually think the third one is three and a half small boxes.

0:14:23 So four small boxes would be 0.16. Three and a half would be 0.14. Let's count the next one. The next one is kind of a little tougher because of the swoop here. See the swoop that I'm referring to? This big swoop. So where does it start? So I know where the P wave starts. It starts halfway in between right where the cursor is. Where is the actual R wave? Because the upslope starts sooner. Right? So I would say probably about here is where I would consider that. So I would measure from here to right there. One, two and a half. Two and a half small boxes.

0:15:10 So I would think that this fourth one is 0.10 at two and a half small boxes. I think the next one is going to be the same thing. I think it's going to be two and a half small boxes. Can I zoom in? Zoom in. Oh, no. That's not what I wanted to do.

0:15:47 So it starts halfway in between. And I think it starts really elevating right here at this line right there. So one, two, and then half of that box. Same thing with the next one, right? Starts right there. So one, two, and that P wave starts just after that line. Two and a half boxes.

0:16:17 And then the last one, that P wave probably starts right here, halfway in between. And the R wave starts right there. So one, two, three and a half boxes. So I got four boxes, four boxes, two and a half boxes, two and a half boxes, three and a half boxes. So what would I expect you to chart for the PRI? I would expect you to chart a range.

0:16:44 I would expect you to chart the shortest one to the longest one. So then I would go 0.10-0.16. So you're giving me the shortest one and the longest one for the PRI documentation on the homework and the test. By the way, our next test is in two weeks, and that next test will not be a multiple-choice question test. It's 15 strips. It will look just like your homework assignment, okay? It will include sinus rhythms, atrial rhythms, and junctional rhythms. So if you list out all the rhythms that we're going to talk about, like last week, we talked about normal sinus rhythm. You know out of those 15 strips, there will be one strip that's a normal sinus rhythm.

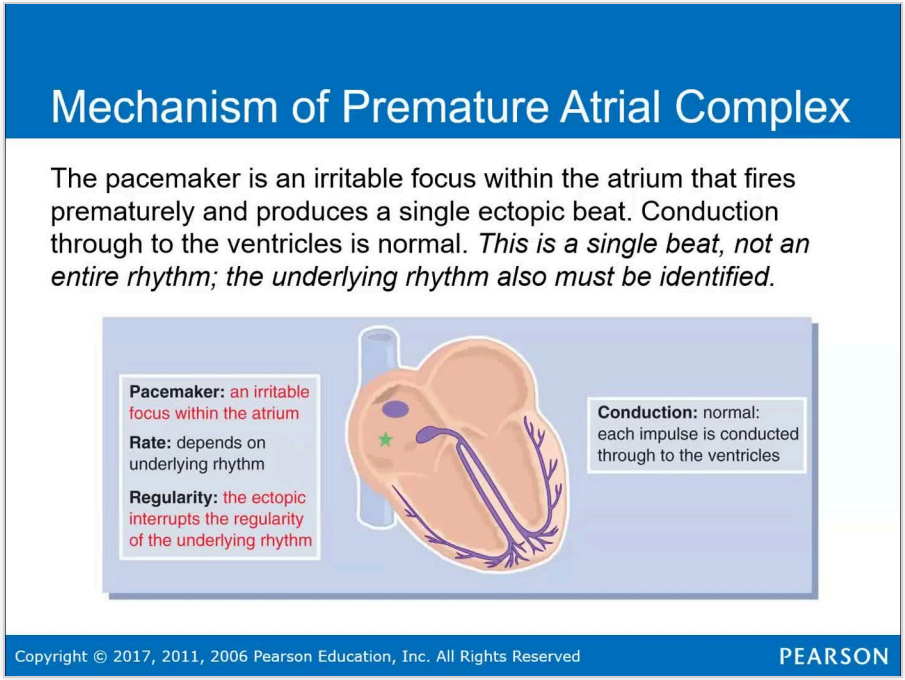
0:17:44 Sinus bradycardia, there will be one strip. Sinus tachycardia, there will be one strip. Sinus arrhythmia, there will be one strip. I told you I'm not going to put Wandering Pacemaker on the test, but we're going to have our next rhythm we're going to talk about is rhythms with a PAC in it. And we're going to talk about atrial fib, atrial flutter, atrial tachycardia. And then we're going to talk about junctional escape, premature PJCs, premature junctional contractions. We're going to talk about accelerated junctional. We're going to talk about junctional tachycardia. So there's going to be one of each of those rhythms on the test. That's 12 rhythms.

0:18:33 And then that's going to leave me three rhythms to choose, right? Whether I want to put another sinus tach on there, or I want to put another junctional tach on there, right? So there'll be at least one of each. And then there will be three more of my choosing that will be there. It will be out of the rhythms that we've gone over in class. Not Wandering Pacemaker. All right, let's measure the QRS. I'll just leave it zoomed in. You can see the QRS starts right here. And it ends right there. So that's going to be one, two, three small boxes. One, two, three small boxes.

0:19:21 And so then our interpretation. So we have a rhythm that is slightly irregular or irregular. We have a heart rate of 60. We have P waves that the shape changes. We have PRIs that vary. We have a normal QRS width. This can only be Wandering Pacemaker. Based on the P wave shape changes and the changes in the PRI, it can only be Wandering Pacemaker. Okay, questions? Okay.

9 Mechanism of Premature Atrial Complex

0:20:01 – 0:21:52



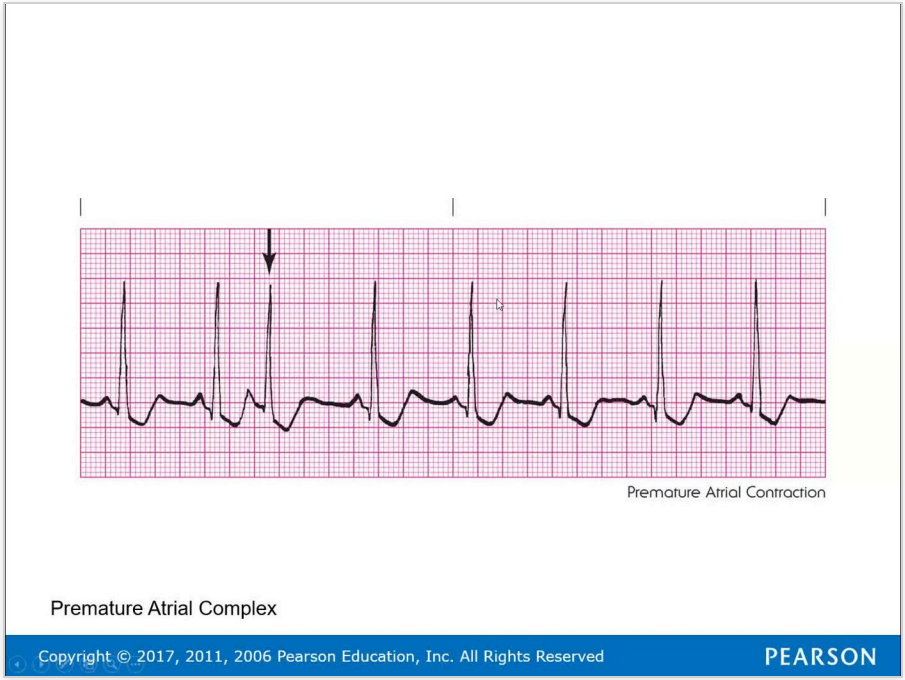
0:20:01 Now we're going to talk about something called a premature atrial complex. A couple weeks ago, I was talking about irritability. When a cell, a cardiac muscle cell, and this could be a muscle cell or it could be one of the nerve cells in the heart. When that cell gets irritable, it can generate an early beat, one single early beat, or if it gets irritable enough, it can fire repetitively.

0:20:45 If the rate at which that irritable pacemaker is faster than the inherent pacemaker, i.e., the SA node, right? So let's say one of the atrial muscle cells right here fires, but it gets irritable, and it fires repetitively, and it fires at a rate faster than the SA node. Remember one of the rules we talked about, fastest rate wins. So it will convert from whatever sinus rhythm they were in to this atrial rhythm that they're now in because this cell fired more than once. It's firing repetitively. You'll see what I mean in a minute, okay? So this irritability causes early beats or premature beats, and that's why we call irritable beats premature beats.

0:21:34 It's an ectopic beat, but the main classification is premature. Why do we call it premature? Because it comes early in the cycle. So I'm just going to move ahead to the rhythm just for a second.

10 Premature Atrial Complex (ECG strip)

0:21:52 – 0:22:48

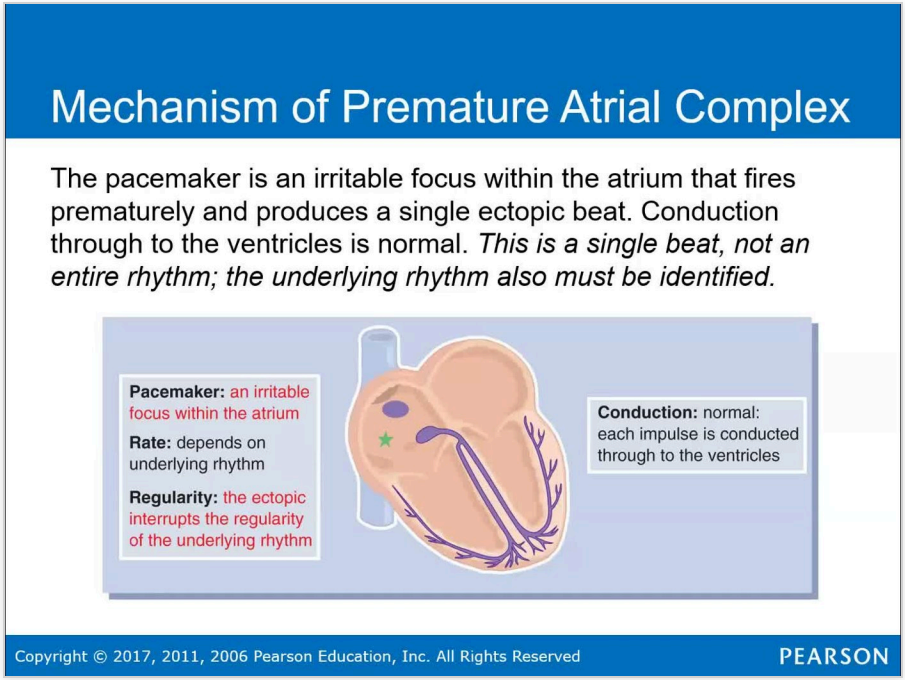


0:21:53 I want you to understand what I'm talking about. So this first beat and second beat is the normal spacing for this particular rhythm. So if I map this out, if I actually map this out, this spacing is the normal spacing. You know how I know that because if I compare this spacing to that spacing to that spacing to that spacing to that spacing, they're pretty much all the same. So what happens here is we have a normal sinus beat followed by a normal sinus beat, and look at the gap here. This gap is short, so this beat comes earlier than the normal cycle would. That's why we call them early beats.

0:22:39 That's why we call them premature beats. So when we have a beat that comes early, it's a premature beat, and it is always from irritability.

11 Mechanism of Premature Atrial Complex

0:22:48 – 0:24:40

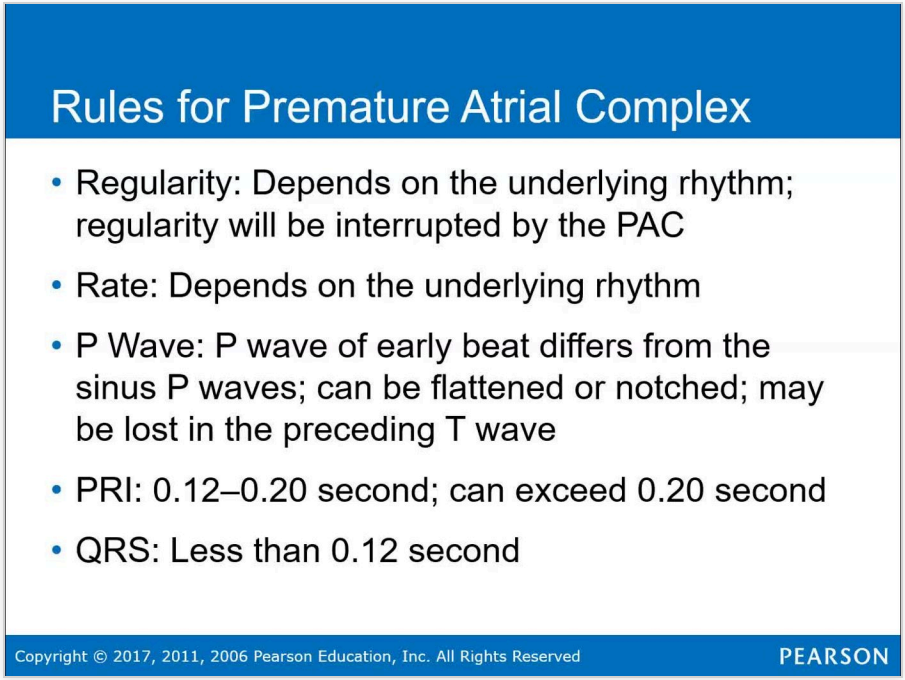


0:22:49 And irritability can be caused by a number of different reasons. It could be caused by the person has coronary arteries disease, and that portion of the heart is not getting enough oxygen. It could be caused by, you know, you ingesting too much caffeine. It could be caused by, you know, it's time to put the crack pipe down and stop smoking or snorting coke, right? Because those are stimulants that's going to make your heart want to jump and go faster, okay? So irritability can be caused by a number of different things. So in this particular rhythm with a premature atrial complex, I'm really talking about a single beat that is a premature atrial complex.

0:23:42 This is occurring inside of an underlying rhythm, so we're going to have to interpret A, the underlying rhythm, and we're going to have to interpret the premature beat in our interpretation, okay? So the pacemaker for this premature atrial complex, also known as a PAC, is an irritable focal point within the atria that fires prematurely and produces a single ectopic beat. During the single ectopic beat, conduction through to the ventricles is normal. Again, because the beat's coming from above the ventricles, we're going to have a narrow QRS. So this is a single beat, not an entire rhythm. We still need to identify the underlying rhythm. Let's go to the guidelines, and then we're going to go to interpreting the rhythm.

12 Rules for Premature Atrial Complex

0:24:40 – 0:25:43

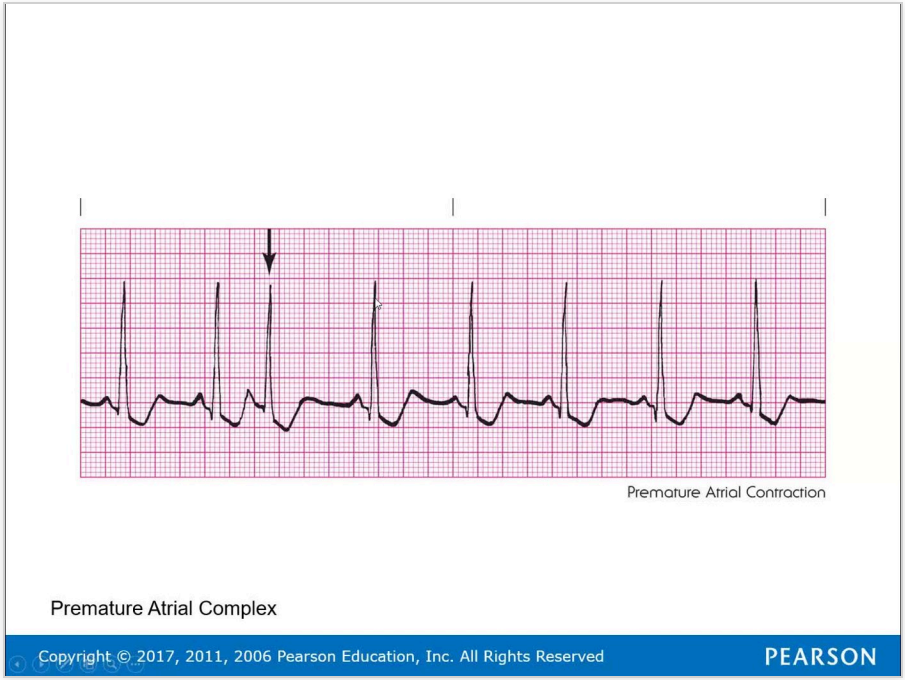


0:24:45 So the regularity for this particular rhythm is going to depend on the underlying rhythm, not the premature beat. With one atopic, right? Because I could give you a rhythm with one PAC in it. I could give you a rhythm with two PACs in it. I could give you a rhythm with three PACs in it. And I'm going to explain, when we get to the rhythm, I'm going to explain how do we determine whether the underlying rhythm is regular or not. Besides the premature beats, how do we determine whether it's regular or not, okay? And it's basically going to be by the R to R intervals.

0:25:33 Again, we're going to count out the R to R intervals across the top of the rhythm string, right? And you know where I showed you where we had the early beat, and then after we have an early beat, we have a compensatory pause.

13 Premature Atrial Complex (ECG strip)

0:25:43 – 0:37:07



0:25:49 That's normal. So you can see that the gap shortened, the gap lengthened, and then we went back into the normal spacing, okay? We'll get there momentarily. I went in the wrong direction. So the underlying rhythm is going to be based on R to R, whether it's regular or not. The rate, again, is going to depend on the underlying rhythm. And the same rules apply. If the rhythm is regular, we're going to divide the spacing between R to R into 1500. If the rhythm is irregular, count the number of R's and multiply by 10. So those rules are not going to change. The P wave. Now, the P wave for this PAC will look different than the underlying rhythm's P waves.

0:26:38 Again, this is an atrial beat. Well, I'm going to repeat myself again. Atrial beats are peaked, flattened, or notched. Atrial P waves, peaked, flattened, or notched. Sinus P waves are round and dome-shaped. So the P wave for this premature beat is going to be different than the sinus P waves or the underlying rhythm's P waves. They can be flattened or notched or peaked. And depending on how early it comes, sometimes its P wave is buried in the preceding beats T wave. Remember, when you come early, there's less space between beats. And now some of the waveforms can mush together. You'll see what I mean as we go through this lecture.

0:27:32 The PRI is going to be normal. And the QRS will be narrow. So let's go through our information gathering. So the first thing we're going to do is count number of small boxes between ours. 5, 10, 15, 16, 17, 18, and two halves make 19. So I have 19 small boxes between the first two R waves.

0:28:02 5, 6, 7, 8, 9, either 10 or 10 and a half. I'll call 10 and a half between the second and third R waves.

0:28:15 5, 10, 15, 15, 19, 20, 21. I'm going to call 21 between the third and the fourth. 5, 10, 15, 15, 4 is 19 and a half. So 19 and a half between the fourth and the fifth. 5, 10, 15, 16, 17, 18, 19.

0:28:49 5, 10, 15, 10, 15, 15, plus 4 is 19. So I got 19, 10 and a half, 21, 19 and a half, 19, 19, 19, 19. So here's the rule of thumb for regularity. As long as you can find two sets of R to R that are exactly the same width apart, the underlying rhythm is regular. Do you want me to repeat that?

0:29:30 In order to determine if an underlying rhythm is regular, you have to find two sets of R to R that have the same measurement. Again, within a box and a half of each other. So let's say I gave you a bunch of premature beats all the way through here. But this is 19 and this is 19. And the rest of this is not 19. The underlying rhythm is regular because there's two different spots that have the same spacing. Then we just have to identify what's going on in the middle. In this particular case, we only have two spots here that are not 19. And that's where before and after the premature beat.

0:30:24 Everybody can see how that's an early beat, right? Everybody understand that concept of an early or premature beat? Okay. So then we're going to go 1500 divided by 19. Somebody already do that math. 78.

0:30:45 So the rig, so regularity. Let me take a step back. So the regularity. How are we going to list this? So the regularity. It's going to be regular with. See where the line over it represents with. One ectopic.

0:31:07 Or you can order to do a W slash regular with one ectopic. I'm going to show you that. I'm going to show you that. Okay. So then our heart rate is going to be 78. It's our next piece of information. P waves. Do all of these P waves look the same?

0:31:50 Except for the third P wave. It looks different. So if you look at that third beat. So, you know, I always think you should orient yourself. So you should know what you're looking for. So if I'm looking at that first beat, I see my P wave. I see my QRS followed by a little ST depression here, followed by my T wave right here. Okay. And then I have a P wave with my QRS, a little ST depression. Now, I actually have a combination T wave, P wave here. If you look at the previous T wave, it's not really peaked. It's kind of rounded. So the P wave for this third beat is right on top of the T wave for the second beat.

0:32:45 And it is peaked. Everybody sees that. It's like a mountaintop, right? It's peaked, right? So I know that this P wave looks different than the other P waves that are there. Does that make sense? Everybody see what I'm talking about? Okay. And then it goes back to looking like the same P wave for the rest of the rhythm.

0:33:08 So for the P waves, I would just put upright and uniform. You don't have to describe that one P wave. What you have to recognize is that P wave is different than the rest of the P waves. And when that P wave is different, what does it mean?

0:33:28 It's coming from somewhere else. It's not coming from the SA note. It's got a peak to it. And I told you, what do atrial P waves look like? They can be flat on the top of the P wave, like, like flat, like literally like a plateau, or it could be notched, like it could go up, have a little hitch down, and then go back up again, right? Like, like a smaller mountain next to a smaller mountain next to a bigger mountain with a little notch in there, right? Or peak, like this particular one.

0:34:02 So P waves are upright and uniform for the underlying rhythm. Now we've got to count our PRI. So I think this PRI is three and a half small boxes, which would be 0.14. By the way, I noticed on the homework and on the test, some of you were giving me boxes. I do not want boxes. I want time. So I'll take it on the homework that you handed in today. But from here on out, I want time. You should know how to measure time with the boxes, okay? Yes, I'm making you do math. Sorry.

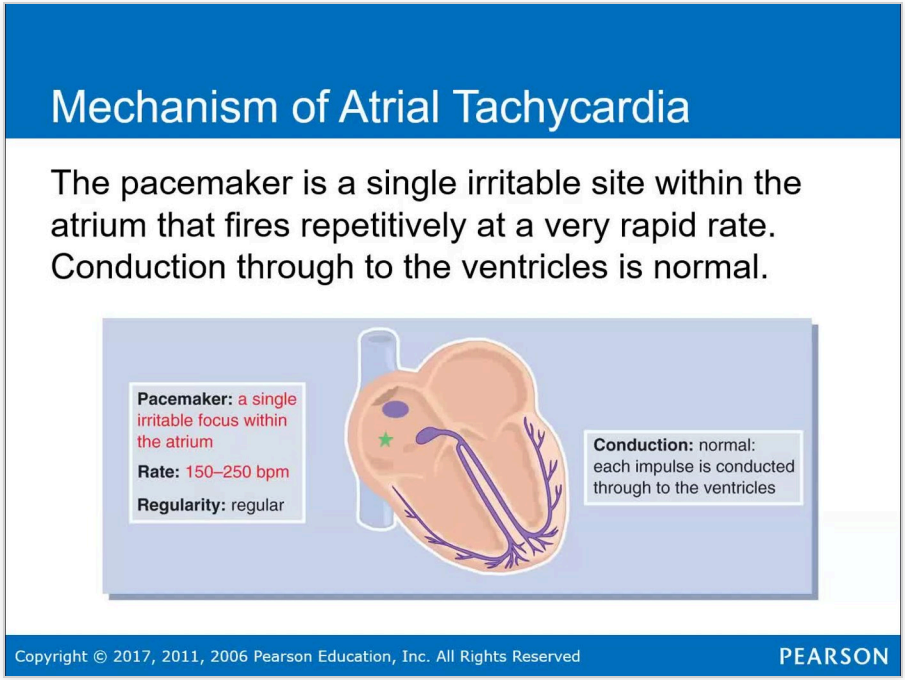
0:35:12 You're going to have to do math for the rest of your life in some way, shape, or form. All right. Now I've got to measure the QRS.

0:35:27 One, two and a half. One, two and a half. So I'm pretty comfortable with that being two and a half, which would be 0.10. Now, how do I put all of this together? What's the underlying rhythm? The underlying rhythm is regular with a heart rate of 78, P waves that are upright and uniform, with a normal PRI and a normal QRS. Where's the underlying rhythm coming from? SA node, right? And everything is normal. So this underlying rhythm has to be NSR, normal sinus rhythm. Now we know we have an early beat because we have a short spacing with a different shaped P wave.

0:36:18 That beat's not coming from the SA node because that P wave is shaped differently. What do junctional P waves look like? They're either A, not visible, or inverted. So the only, I repeat, the only place that this beat can come from is somewhere in the atria. So our interpretation for this rhythm is normal sinus rhythm, NSR, with one PAC, premature atrial complex. Everybody understand? Okay. Okay. All right.

14 Mechanism of Atrial Tachycardia

0:37:07 – 0:38:22

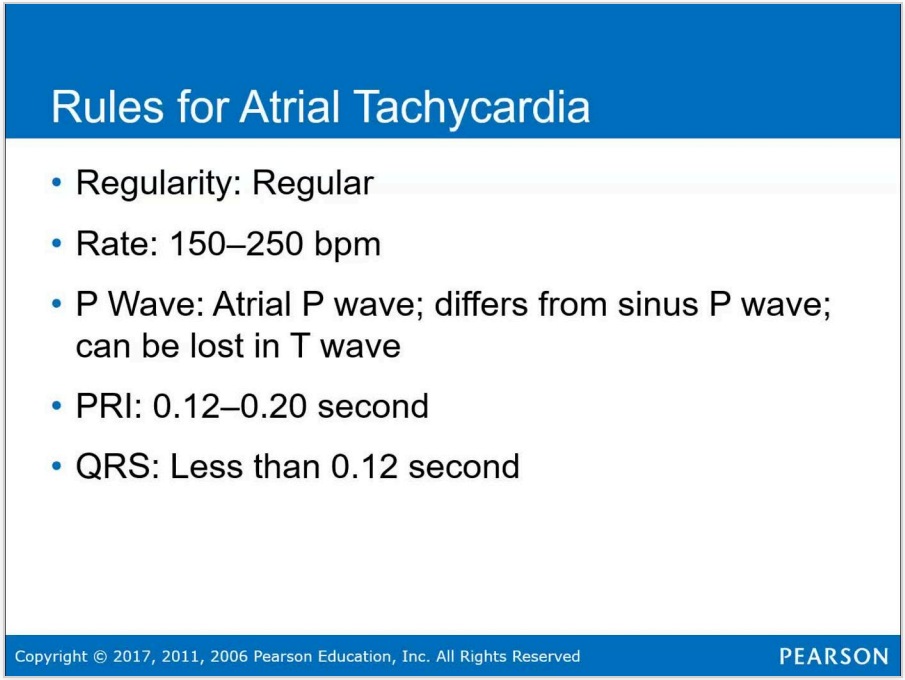


0:37:08 Now I'm going to move on to atrial tachycardia. So this is a single irritable focal point within the atria, just like a PAC. But now, instead of firing just once, it's going to fire, and it's going to fire repetitively at a rate much higher than the SA node can fire. So it's actually going to take over as the pacemaker. So I started this by talking about irritable focal points can cause a single beat, or it can cause an entire rhythm. In this case, it's an entire rhythm because it's firing repetitively at a rate faster than the SA node, and now takes over as the pacemaker.

0:38:06 So pacemaker is a single irritable site within the atria that fires repetitively at a very rapid rate. Conduction through the ventricles is normal. What that's saying is we're going to have a narrow or a normal QRS. So let's go to our guidelines here.

15 Rules for Atrial Tachycardia

0:38:22 – 0:41:22



0:38:23 R-to-R spacing for atrial tachycardia will be regular. I said a narrow or normal QRS. Most of the time, we will not be able to identify P waves in this rhythm.

0:38:49 This is, so, you know, I give you a seven or eight beat in the heart rate calculation. I give you a seven or eight beat grace period, as long as your rate is within seven or eight beats of the rate that I calculated. And you'll see on the homework that I handed back today, when I have a different measurement than you, I write it down next to your answer, but I put it in parentheses. So you'll see it in red ink in parentheses. So you know what my measurement is compared to your measurement. It doesn't mean I took points off. If I take points off, you'll see a minus one or a minus two or a minus three.

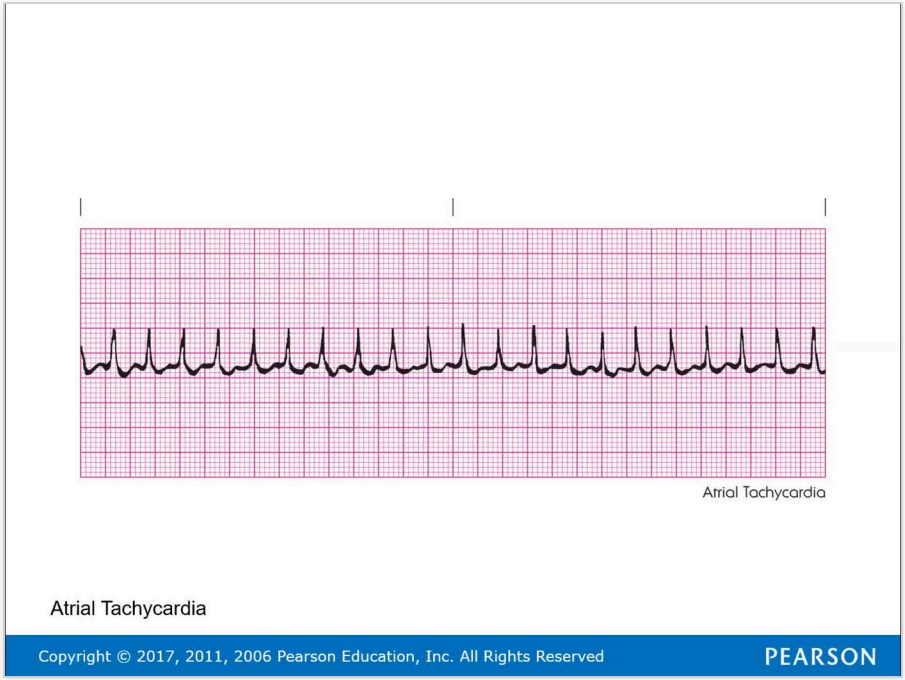
0:39:29 This is important that you get the spacing between R-waves correct, because when the rate goes this fast, your margin of error shrinks. So let's say R-waves are seven boxes apart. If the R-waves are really seven boxes apart, that means the heart rate is 214 beats per minute. If you tell me it's eight boxes apart, that means the heart rate is going to calculate out at 187. And you'll be way outside of that six or seven beat grace period. So you've got to be accurate with the number of small boxes between R-waves. So the faster the rate or the smaller the space between R-waves, the less margin of error you will have in order to get a correct heart rate.

0:40:27 So the rate for this rhythm is usually somewhere between 150 and 250 beats per minute. It is fast. P-waves are atrial P-waves. So they can be flattened, notched, or peaked. Most of the time, we do not see P-waves in this rhythm because it's going so fast, there's not enough space

between beats to see all of the waves. So most of the time, the P-wave will be lost in the preceding beats T-wave. And if we don't see the P-wave, we cannot measure the PRI. We have to see the P-wave in order to measure the PRI. And then the QRS will be narrow, and this is what it looks like.

16 Atrial Tachycardia (ECG strip)

0:41:22 – 0:48:50



0:41:27 Now, the first thing I want to talk about when you see a rhythm like this. It's really fast. Is this a life-threatening rhythm? If somebody goes into this rhythm, is their life in danger?

0:41:46 Why? Why are we taking down the stroke line? That's the key to this answer. So you don't have enough filling time, ventricular filling time, right? So the faster you go, the less time the ventricle has to fill with blood. Remember, cardiac output. What's the formula for cardiac output? Heart rate times stroke volume. We have plenty of heart rate. What we don't have here is stroke volume. So each time the heart contracts, it only squeezes out a tiny amount of blood. So the overall cardiac output in this rhythm actually drops, even though we have an exceedingly amount of heart rate. So this is a life-threatening rhythm.

0:42:46 This patient will probably be hypotensive and dizzy or short of breath, pale, cold and clammy, maybe not even a good mental state, altered mental state. Your question. That is a tachycardia coming from the ventricles. We're going to get there in a month. All right? So next time we'll have our test. And then actually I think I cover ventricular rhythms the second half of class. So we'll get there in two weeks, actually, for VTAC. We're going to do that today. We're going to do that today. All right. So let's go through our information gathering. One, two, three, four, five, six.

0:43:49 I think that's seven. Seven small boxes. Five, six, seven. Four, five, six, six and a half. Seven and a half. Seven, seven, seven, seven. So I'm pretty comfortable that the average space between our waves is seven. So this rhythm is regular.

0:44:27 Heart rate is 214 beats per minute. Heart rate is 214 beats per minute. Again, orienting yourself. This is not a P wave. I can't see all of this beat, but this is T wave. Most likely T wave, P wave combination. Here's my QRS. T wave, QRS, T wave, QRS, T wave. There are no visible P waves here. So under the P waves you rate, not visible.

0:45:05 P, R, I can't measure. QRS. One, two. One, two. One, two. I'm pretty comfortable that this QRS is 0.08 seconds. So it's actually hard to interpret this rhythm. It's hard to interpret this rhythm because a lot of times we have to identify P waves in order to identify where the rhythm is coming from. And we don't have any P waves in this rhythm.

0:46:01 So as I get to the end of this lecture, into the beginning of the next lecture, there's a catch-all phrase that we use for rhythms. Narrow complex tachycardias. So when I say narrow complex tachycardia, that means it has a narrow QRS.

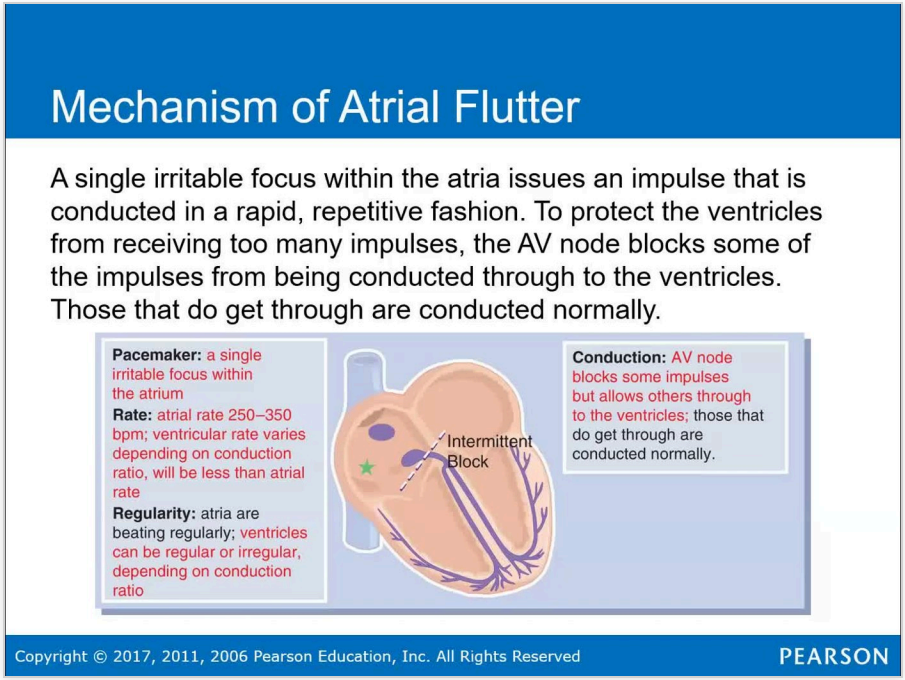
0:46:26 Where do all narrow beats or rhythms come from? Either the SA node, an atrial rhythm, or a junctional rhythm. Because the narrow QRS tells me this origin for this rhythm is coming from above the ventricles. What prefix do we use for something that says coming from above? Supra. So this, so I could say narrow complex tachycardia, or I could say supraventricular tachycardia.

0:47:10 Or SVT, supraventricular tachycardia. Now, this catch-all phrase of supraventricular tachycardia is used for narrow complex tachycardias that have rate, heart rate ranges similar to other rhythms, but we can't differentiate because we cannot see the P waves. So SVT is a catch-all phrase interpretation that we use for narrow complex tachycardias that have similar rate ranges to other narrow complex tachycardias, but we cannot differentiate one from the other because we cannot identify the P wave. That's a lot in one sentence, but it is what it is. I don't make the rules. I just teach the rules. Okay? So, if you, on this interpretation, if you told me this was atrial tachycardia, I wouldn't mark it wrong.

0:48:25 Because this is one of the very few rhythms, this is one of two rhythms, that actually gets above 200 in adults. But most of the time, what I want you to put for this interpretation is SVT. And you'll get a little better understanding of SVT as I move forward with some of the other rhythm interpretations. Okay? All right. I'm going to keep moving.

17 Mechanism of Atrial Flutter

0:48:50 – 0:52:01



0:48:50 So, next rhythm we're going to talk about is atrial flutter. Like atrial tachycardia, this is a single irritable focal point within the atria that fires at a rapid, repetitive fashion. Now, one of the jobs of the AV node is to limit the number of impulses that are sent down to the ventricles.

0:49:37 So, in this particular case, the AV node blocks some of the impulses from this atrial focal point that is firing in a rapid, repetitive fashion.

0:49:54 So, for example, this atrial focal point fires, the AV node blocks it. What do I mean by block? It doesn't send the impulse down for ventricular depolarization. So, it fires, the AV node blocks it. It fires again, the AV node sends it down for ventricular depolarization. It fires again, it blocks it. It fires again, it sends it down. So, what I was giving you is the pattern of block one, depolarize one. Block one, depolarize one. Okay?

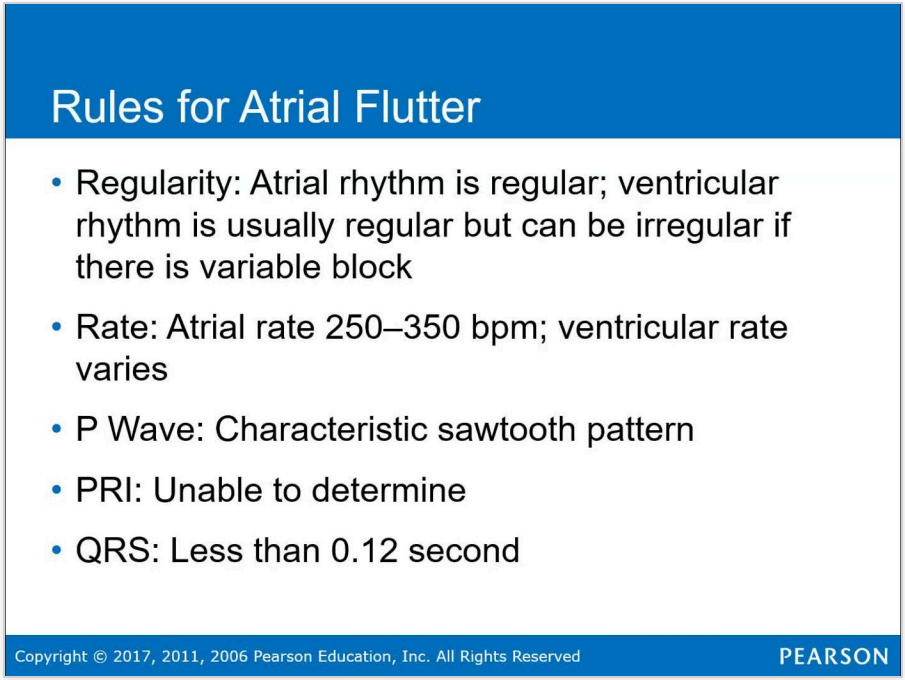
0:50:28 In atrial tachycardia, the AV node's not doing its job. It sends every impulse down. It sends every impulse down for ventricular depolarization. So, atrial tachycardia and atrial flutter are really the same irritable focal point, except in atrial flutter, the AV node is doing its job by blocking some of the impulses, limiting the ventricular radius.

0:51:04 So, to protect the ventricles from receiving too many impulses, the AV node blocks some of the impulses from being conducted through the ventricles. The ones that are sent down will be conducted normally, meaning the QRS will be normal or narrow. So, again, this is a single irritable focal point that fires at a very rapid, repetitive fashion. In fact, on average, for this

rhythm, that focal point is firing somewhere between 250 and 350 impulses per minute. So, the atrial rate is going to be pretty fast, but the ventricular rate is not going to be as fast because the AV node is blocking some of those impulses. Okay? Okay? So, rules.

18 Rules for Atrial Flutter

0:52:01 – 0:56:06



0:52:08 And I'm not going to actually have you assess for atrial rhythm regularity. I'm only going to have you assess ventricular rhythm regularity. So, usually, in atrial flutter, R-to-R spacing will be regular. There are times where R-to-R spacing is irregular in this rhythm. This is where we're starting to get exceptions on top of exceptions. So, if the AV node has a consistent block, and I gave you an example of what I mean. It blocked one, conducted one. Blocked one, conducted one. So, that means it's blocking every other impulse and conducting every other impulse. So, as long as the AV node block ratio stays consistent, and in this case, blocks one, conducts one, blocks one, conducts one, I would call that a two-to-one ratio.

0:53:29 It's blocking every other, conducting every other, so, for every two times that it beats, it blocks one, conducts one, okay? Now, the AV node does not always do that in the same fashion. So, what if the AV node blocked one, blocked another, and then conducted one? And then blocked one, conducted one. So, it blocked two, and then conducted one. And then it blocked one, and then conducted one. So, if you think about the timing, the timing of the block is going to dictate R-to-R space. If I block two here, and then conduct one, R-to-R is going to be farther apart, because I blocked one, blocked one, conducted one.

0:54:26 And then on the next one, I only blocked one, and then I conducted one. Those R waves are going to be closer together. So, when the block ratio stays the same, R-to-R will be regular. When the block ratio varies, R-to-R will be irregular. So, that's why it's saying ventricularism.

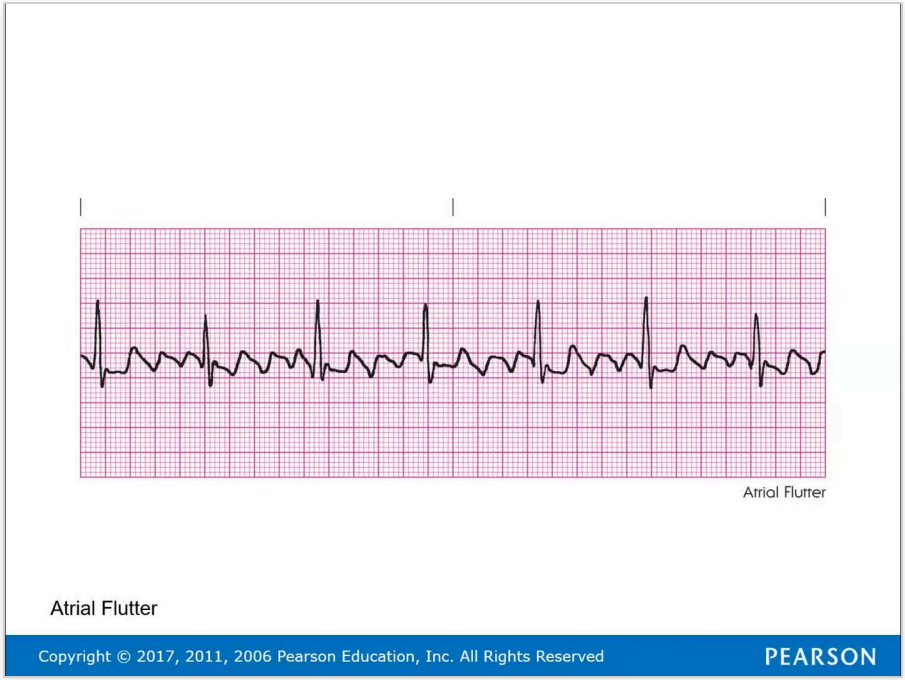
The ventricular, or R-to-R, is usually regular, but can be irregular if AV node's block is variable.

0:55:03 There is an easy, identifiable feature for this rhythm. I'm going to share that with you in a second. The atrial rate's usually going to be pretty fast. The ventricular rate is usually going to vary. It's usually within the normal range, because AV node is doing a pretty good job. Here is the identifiable feature of this rhythm. The P-wave. It shows up characteristically looking like a sawtooth pattern. Have you ever seen a handsaw and the teeth on a handsaw? And they're jagged. They're like triangular and jagged, right? Put an asterisk next to this P-wave, because any time you see the P-waves appearing in a sawtooth pattern, it can only be atrial flutter.

0:55:54 We cannot measure the P-R-I in this rhythm, and the Q-R-S will be narrow.

19 Atrial Flutter (ECG strip)

0:56:06 – 0:59:46



0:56:08 And this is what it looks like. You should easily be able to see what looks like teeth on a saw blade. And let me clue you in what's happening here. So this, I don't know what's happening before this Q-R-S, but this first tooth on the saw blade got blocked. It didn't get conducted. I know it got blocked because it didn't, a Q-R-S didn't follow it. Everybody understand that? If you see a P-wave not followed by a Q-R-S, that P-wave got blocked.

0:56:49 The second tooth on this saw blade also got blocked. The third tooth got conducted because it's followed by a Q-R-S. So blocked, blocked, conducted. And that pattern stays consistent across this strip. Blocked, blocked, conducted. Blocked, blocked, conducted. So I would look at this as a three-to-one ratio.

0:57:18 For every three P-waves, only one gets conducted. So where is your T-wave? You sort of see it on the first one, but I just don't know. You don't really see much of a T-wave here. Don't really see much of a T-wave. If it is, it's inverted here. So T-waves can be upside down as well.

0:57:47 Okay? All right, let's go through our information gathering here. 5, 10, 15, 20, 21 and a half. Possibly 22 because that one is a little off-center.

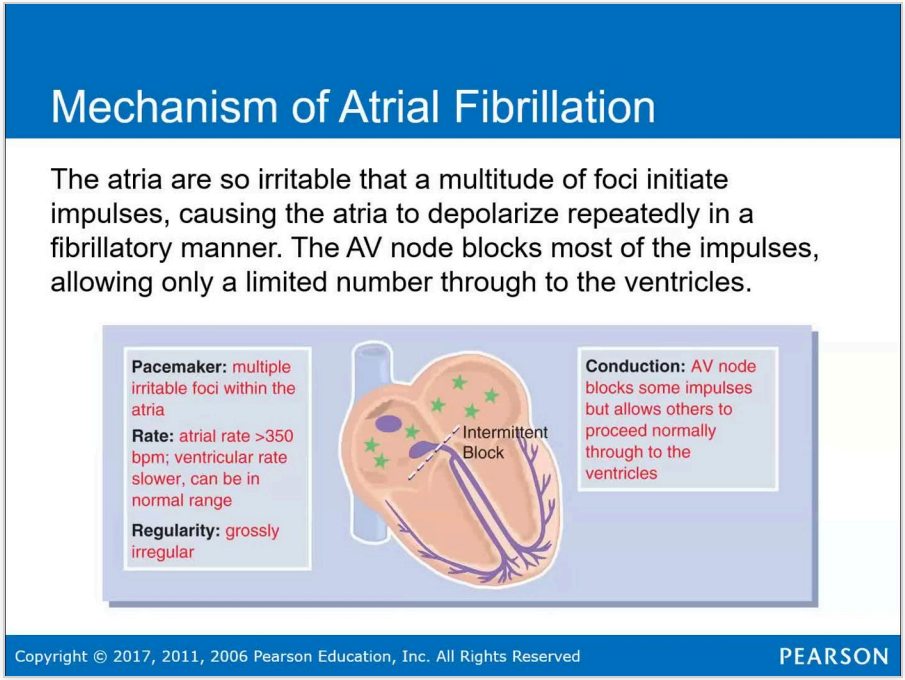
0:58:04 5, 10, 15, 20, 22 and a half. 5, 10, 15, 19, 20, 21 and two halves make 22. 5, 10, 15, 20, 22 and a half. 5, 10, 15, 19, 20, 21, 22. So this rhythm is regular. I knew it was going to be regular because the ratio stays three-to-one across the strip.

0:58:39 Heart rate is going to be 1,500 divided by 22. 68. So heart rate is 68. P-waves. What do I put for the appearance of the P-waves?

0:59:01 Sawtooth. That's what I'm expecting you to put for the P-wave appearance. P-R-I? Can't measure. Q-R-S? One, two and a half. One, two and a half. Yeah, I'm pretty convinced it's two and a half small boxes at 0.10 seconds. And because I have this sawtooth appearance, my interpretation is atrial flutter. All right, I'm going to keep moving.

20 Mechanism of Atrial Fibrillation

0:59:46 – 1:04:32



0:59:47 Our next rhythm here is called atrial fibrillation. So this is not one single irritable focal point. It is many irritable focal points. Or in this case, we would say multiple foci, meaning multiple different focal points. And you can see all of the different green stars, right? So all of those green stars are creating the impulses that are not timed, that are not coordinated. They're not working in unison with each other. So the atria have become so irritable that we have multiple focal points initiating impulses at the same time.

1:00:44 So the cardiac muscle cells cannot respond appropriately to multiple impulses at the same time. So this causes the atrium muscle cells to depolarize repeatedly, not in a manner in which the cells actually contract. What they do is they fibrillate. So there's no actual atrial contraction here. It's like the atria are just quivering, trying to respond to every single impulse, which is impossible.

1:01:22 So in this rhythm, we lose atrial contraction. Remember, the heart's meant to beat from the top down. I want the atria to contract before the ventricles, because I want to maximize the amount of blood in the ventricles prior to ventricular contraction. So in this case, we don't have atrial contraction, so we lose some ventricular filling in this rhythm. Physiologically, the stroke volume gets smaller in this rhythm because there's no atrial contraction.

1:02:08 The other thing that can happen to blood inside the atria, because the atria is not contracting, it's not squeezing the blood. So some of the blood can actually get stagnant inside the atria. What happens to blood when it gets stagnant? What happens to water in the winter if it's not

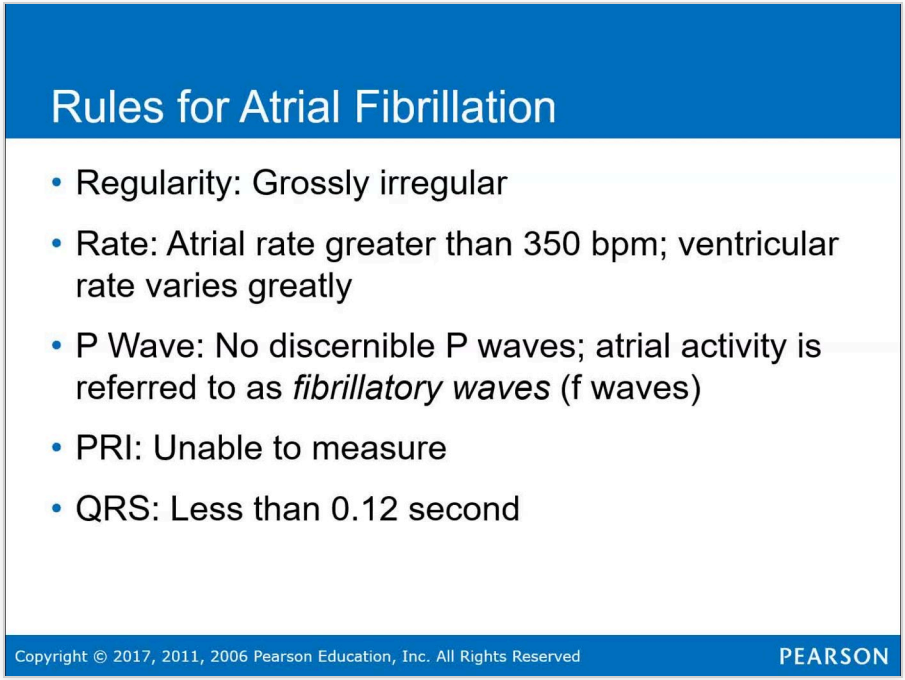
moving? What happens to the top layer? It freezes. What happens to blood when it's not moving? It clots, right? So people who have chronic a-fib are at risk of developing blood clots inside the atria. And if that blood clot gets moved out of the heart, the patient is at risk of that blood clot going up to the brain.

1:02:57 What do we call that? Blood clot to the brain. Stroke. That's called a stroke. So patients who have chronic a-fib are usually on a blood thinner, like Eliquis or Pradaxa. You see commercials for these drugs on TV for a-fib. I think the most recent person on there is Kareem Abdul-Jabbar for a-fib, right?

1:03:31 Okay. So the AV node does do a pretty good job in this rhythm as well. It blocks a lot of the impulses, allowing only a limited number to go through for ventricular depolarization. And we really want to limit the heart rate for these patients. These patients are also put on drugs, atrial arrhythmia drugs, to minimize the heart rate. Because if this a-fib heart rate gets up into the 140s and 150s, remember, we don't have atrial contraction, so the stroke volume gets smaller. But now if the rate gets too high, now we have less filling time, the stroke volume gets even smaller. So we really try to medicate these patients to keep their ventricular heart rate less than 100, because they can become hypotensive if the heart rate gets going too fast.

21 Rules for Atrial Fibrillation

1:04:32 – 1:06:09

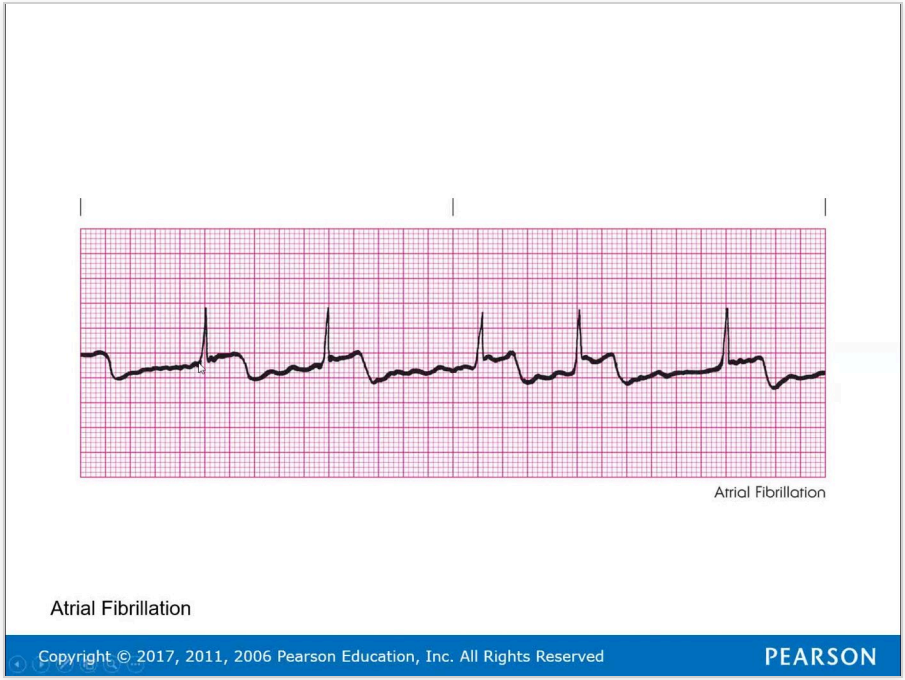


1:04:32 All right. So R to R. So because all of these focal points are firing at the same time, the AV node blocks some, but doesn't block some of the other ones, there's no rhyme or reason to R to R. So we call this grossly irregular. So R to R is usually completely irregular across the entire strip. Atrial rate, again, we can't count looking at an EKG, so I don't expect you to do that. But the ventricular rate is going to vary, depending on the effectiveness of the patient's AV node and how many impulses it blocks. Because if you look back on the previous slide, the impulses are usually greater than 350 from the atria.

1:05:34 So that's extremely fast. There will not be any identifiable P waves in this rhythm. This is also an identifiable feature. What we will see is what looks like artifact at the isoelectric line, because the atria are constantly fibrillating. So you're not going to see P waves. We're going to see what we call F waves or fibrillatory waves. You'll see what I mean in a minute. Can't measure the PRI and the QRS will be normal or narrow, 0.12 or less.

22 Atrial Fibrillation (ECG strip)

1:06:09 – 1:08:01

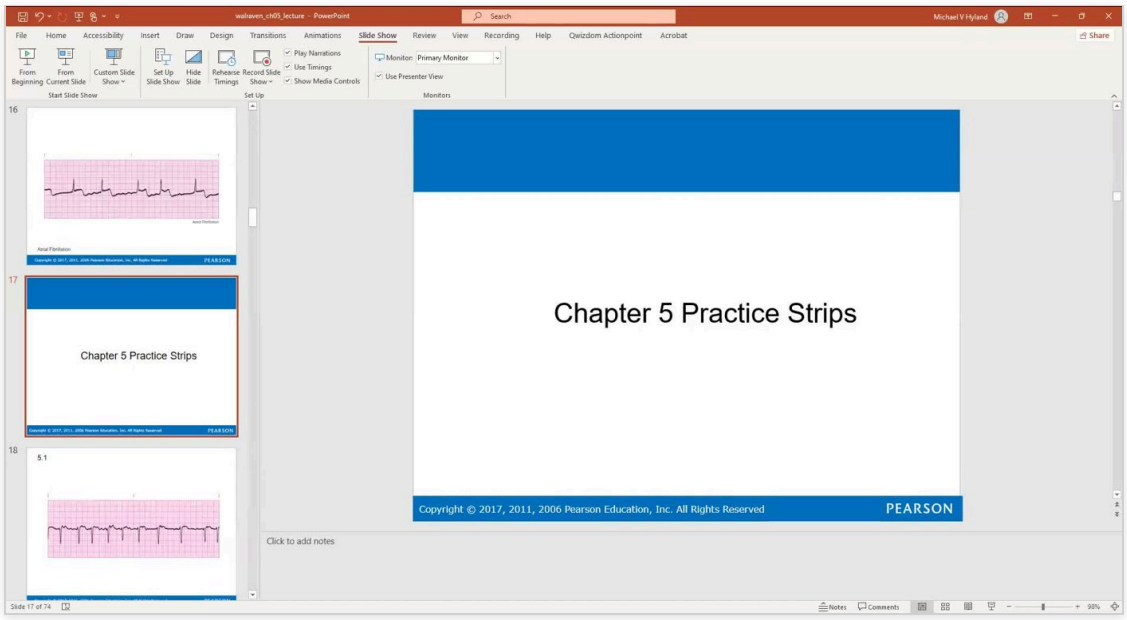


1:06:09 And this is what an example of what AFib looks like. So you can just eyeball this and see that the spacing is completely different across the strip. Right? But let's go through our information. 5, 10, 15, 20, 25 between first and second. 5, 10, 15, 20, 25, 30, 31 between second and third. 5, 10, 15, 19 and a half between third and fourth. 5, 10, 15, 20, 25, 29 and a half between the last two. So grossly irregular. So you don't have to put grossly irregular. You can just put a regular for the regularity. Heart rate. What's the heart rate? 50. It's irregular. Number of R waves times 10.

1:07:09 Now, P waves. You see the undulations are what looks like artifact. It's this constant undulation. Those are fibrillatory waves. So when you see this amount of irregularity with fibrillatory waves, it's A-fib. So for P waves, you can either put F waves or you can put no P waves. I'll take either one. PRI, we cannot measure. QRS, this looks like a pretty narrow QRS. It's a box and a half. 0.06. Interpretation, A-fib. Questions?

23 Chapter 5 Practice Strips

1:08:01 – 1:08:30



1:08:15 All right. I'm going to move into junctional rhythms now.